Chmod cheatsheet

This quick reference cheat sheet provides a brief overview of file permissions, and the operation of the chmod command

Getting Started

Syntax

```
$ chmod [options] <permissions> <file>
```

Example

```
$ chmod 755 foo.txt
$ chmod +x quickref.py
$ chmod u-x quickref.py
$ chmod u=rwx,g=rx,o= quickref.sh
```

Change files and directories recursively

```
$ chmod -R 755 my_directory
```

The `chmod` command stands for "change mode"

Chmod Generator

Permissions: 777 rwxrwxrwx

	User	Group	Other
Read	☑	☑	☑
Write	☑	☑	☑
Execute	☑	☑	☑

Chmod Generator allows you to quickly and visually generate permissions in numerical and symbolic.

Common Permissions

400	r--------	Readable by owner only
500	r-x------	Avoid Changing
600	rw-------	Changeable by user
644	rw-r--r--	Read and change by user
660	rw-rw----	Changeable by user and group
700	rwx------	Only user has full access
755	rwxr-xr-x	Only changeable by user
775	rwxrwxr-x	Sharing mode for a group
777	rwxrwxrwx	Everybody can do everything

Explains

```
$ ls -l
-rw-r--r--  1 root root 3 Jun 29 15:35 a.log
drwxr-xr-x  2 root root 2 Jun 30 18:06 dir
```

Permission analysis of "dir"

```
d  rwx  r-x  r-x
┬  ─┬─  ─┬─  ─┬─
│   │    │    │
│   │    │    └─ 4. Other | 5 (4+0+1)
│   │    └────── 3. Group | 5 (4+0+1)
│   └─────────── 2. User  | 7 (4+2+1)
└─────────────── 1. File Type | directory
```

Permission Modes

Permission	Description	Octal	Decimal
`---`	No Permission	000	0 (0+0+0)
`--x`	Execute	001	1 (0+0+1)
`-w-`	Write	010	2 (0+2+0)
`-wx`	Execute and Write	011	3 (0+2+1)
`r--`	Read	100	4 (4+0+0)
`r-x`	Read and Execute	101	5 (4+0+1)
`rw-`	Read and Write	110	6 (4+2+0)
`rwx`	Read, Write and Execute	111	7 (4+2+1)

Objects

Who (abbr.)	Meaning
u	User
g	Group
o	Others

Who (abbr.)	Meaning
a	All, same as ugo

Permissions

Abbreviation	Permission	Value
r	Read	4
w	Write	2
x	Execute	1
-	No permission	0

File Types

Abbreviation	File Type
d	Directory
-	Regular file
l	Symbolic Link

Chmod Examples

Operators

+	Add
-	Remove
=	Set

chmod 600

```
$ chmod 600 example.txt
$ chmod u=rw,g=,o= example.txt
$ chmod a+rwx,u-x,g-rwx,o-rwx example.txt
```

chmod 664

```
$ chmod 664 example.txt
$ chmod u=rw,g=rw,o=r example.txt
$ chmod a+rwx,u-x,g-x,o-wx example.txt
```

chmod 777

```
$ chmod 777 example.txt
$ chmod u=rwx,g=rwx,o=rwx example.txt
$ chmod a=rwx example.txt
```

Symbolic mode

Deny execute permission to everyone.

```
$ chmod a-x chmodExampleFile.txt
```

Allow read permission to everyone.

```
$ chmod a+r chmodExampleFile.txt
```

Make a file readable and writable by the group and others.

```
$ chmod go+rw chmodExampleFile.txt
```

Make a shell script executable by the user/owner.

```
$ chmod u+x chmodExampleScript.sh
```

Allow everyone to read, write, and execute the file and turn on the set group-ID.

```
$ chmod =rwx,g+s chmodExampleScript.sh
```

Removing Permissions

In order to remove read write permissions given to a file, use the following syntax:

```
$ chmod o-rw example.txt
```

For our file example.txt, we can remove read write permissions using chmod for group by running the following command:

```
$ chmod  g-rx example.txt
```

To remove chmod read write permissions from the group while adding read write permission to public/others, we can use the following command:

```
$ chmod g-rx, o+rx example.txt
```

But, if you wish to remove all permissions for group and others, you can do so using the go= instead:

```
$ chmod go= example.txt
```

Executable

```
$ chmod +x ~/example.py
$ chmod u+x ~/example.py
$ chmod a+x ~/example.py
```

chmod 754

```
$ chmod 754 foo.sh
$ chmod u=rwx,g=rx,o=r foo.sh
```

Chmod Practices

SSH Permissions

```
$ chmod 700 ~/.ssh
$ chmod 600 ~/.ssh/authorized_keys
$ chmod 600 ~/.ssh/id_rsa
$ chmod 600 ~/.ssh/id_rsa.pub
```

```
$ chmod 400 /path/to/access_key.pem
```

Web Permissions

```
$ chmod -R 644 /var/www/html/
$ chmod 644 .htaccess
$ chmod 644 robots.txt
$ chmod 755 /var/www/uploads/
$ find /var/www/html -type d -exec chmod 755 {} \;
```

Batch Change

```
$ chmod -R 644 /your_path
$ find /path -type d -exec chmod 755 {} \;
$ find /path -type f -exec chmod 644 {} \;
```

See: Command Substitution

Also see

Modify File Permissions with chmod (linode.com)